CheckAR: Smart Car Diagnostics

Task 5: UI Design & Implementation

Group 11

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Target User Analysis

Primary Users – Non-Technical Drivers

Everyday drivers with little mechanical experience. Interface uses **simple language, visual guidance, and clear diagnostic explanations** to minimize complexity.

Secondary Users – Mechanics

Professional mechanics needing **fast preliminary diagnostics**, detailed issue descriptions, maintenance history tracking, and mechanic recommendations.

Administrative Users

Admins manage **user accounts, ML model updates, system monitoring,** and performance analytics. Interface focuses on functionality and efficiency.

App Identity Design

Brand Name Rationale

"Check" = vehicle inspection · **"AR"** = advanced smart technology & augmented assistance. The name communicates innovation, intelligence, and reliability.

Logo Components

- White car silhouette — automotive identity
- Green checkmark — verified, reliable diagnostics
- Dark blue circle — trust and professionalism
- Clean sans-serif typography — modern tech feel

Color Psychology & Visual Design System

Primary Color: Professional Blue (#2563EB)

Selected for trust, technology, stability, and readability. Full palette maps colors to urgency levels:

Blue	Primary actions
Green	Healthy status
Yellow	Warning states
Red	Critical alerts

Typography – Inter Font Family

Selected for mobile readability, modern UI design, and excellent spacing.

Headers	Bold · 24px
Subheaders	Semi-Bold · 18px
Body Text	Regular · 14px
Buttons	Medium · 16px

Grid

Responsive mobile grids with adaptive scaling

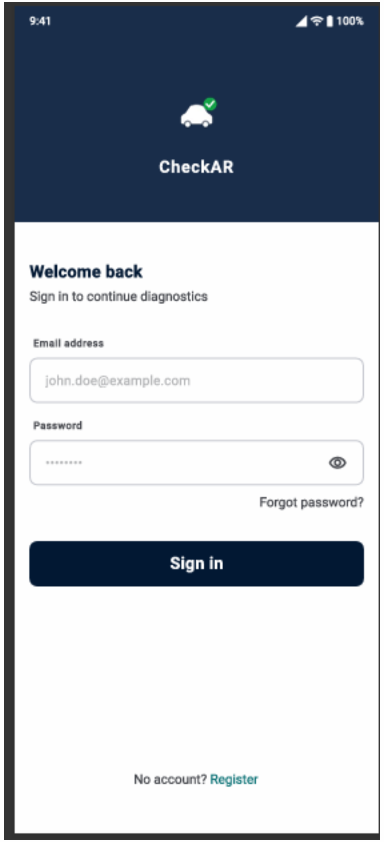
Spacing

4px → 8px → 16px → 24px scale

Icons

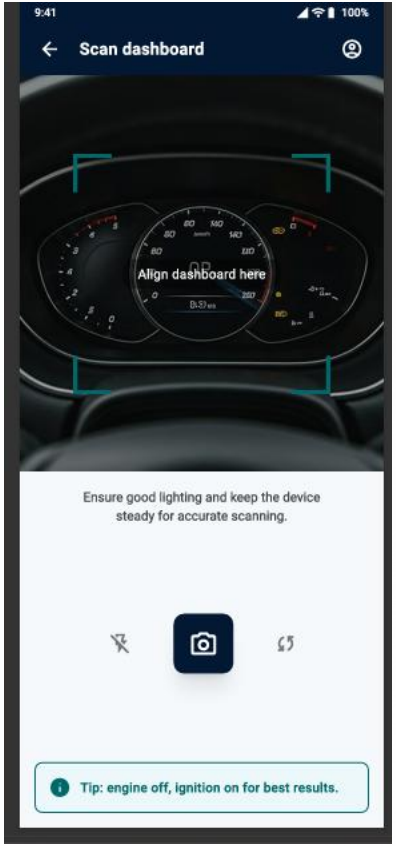
Minimal, scalable, consistent across devices

User Interface Design



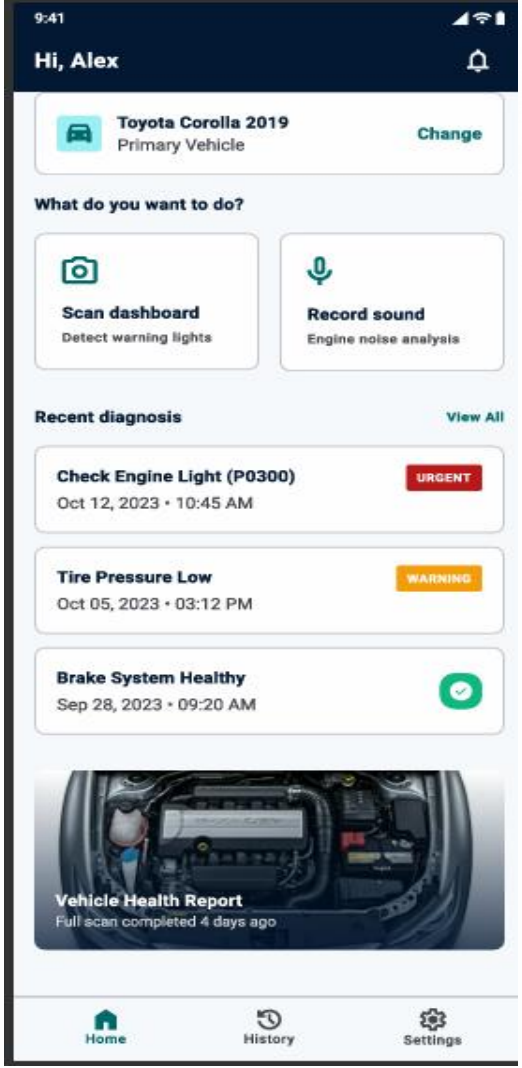
Authentication

Email/password login, registration with role selection (Car Owner / Mechanic), password reset, and input validation with error prevention.



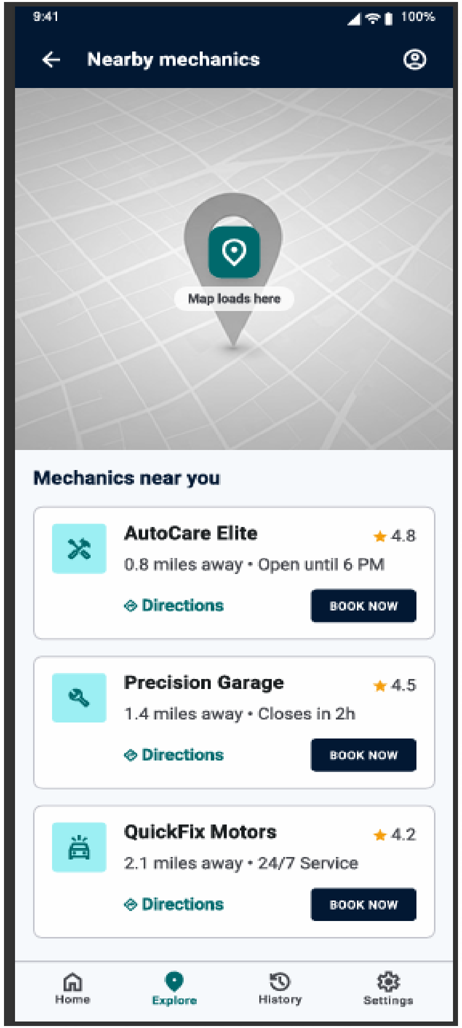
Diagnostic Screens

Camera framing guides, lighting instructions, sound recorder with timer, and results showing severity levels, repair recommendations, and nearby mechanics.



Home Screen

Greeting, vehicle card, diagnostic action buttons (Scan Dashboard, Record Sound), recent diagnosis history, and health summary.



Flutter Frontend Implementation

Why Flutter?

Cross-platform (iOS & Android), fast rendering, rich widget libraries, and accelerated development speed.

State Management

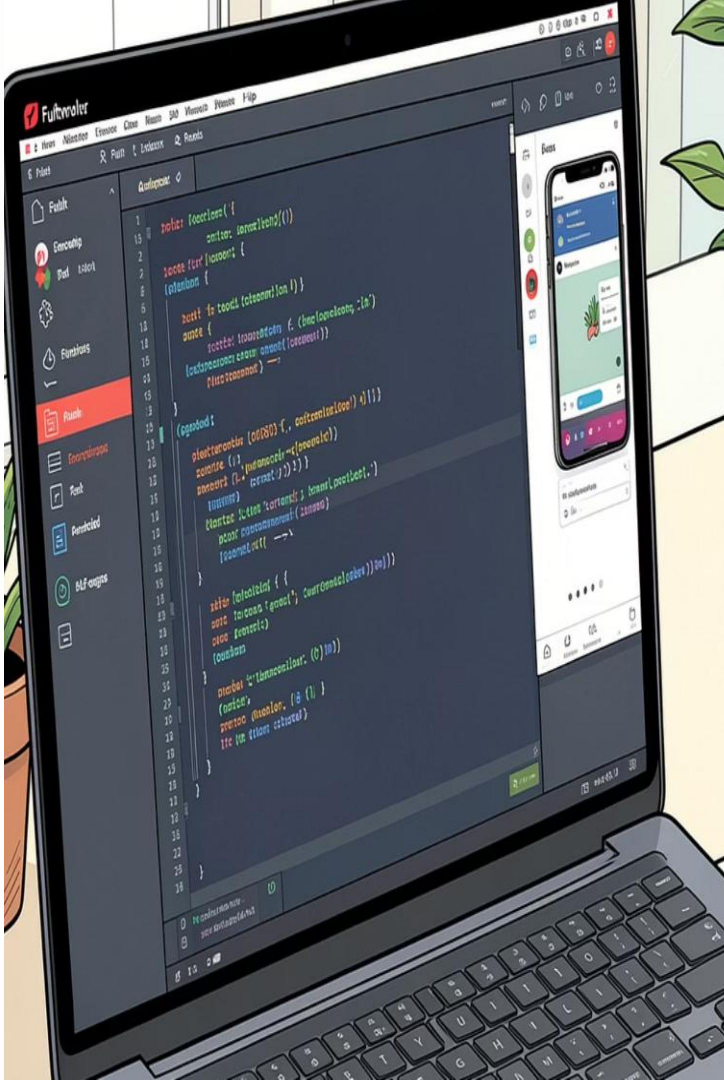
Provider architecture manages auth state, diagnostic results, user preferences, and notifications — improving scalability and maintainability.

Navigation

Bottom navigation bar, named routes, and Flutter Navigator APIs handling page transitions and back-stack behavior.

Key Packages

camera, provider, image_picker, audio_recorder, google_maps_flutter



Screen Implementation Details

01

HomeScreen

Vehicle card, diagnostic actions, recent history, health summary

02

RecordSoundScreen

Microphone permissions, 10-second timer, audio upload progress

03

ScanDashboardScreen

Camera preview, frame alignment overlay, image capture

04

SoundDiagnosisResultScreen

Fault analysis, severity indicators, video tutorials, mechanic locator

05

HelpAboutScreen

App info, support contacts, FAQs, user guidelines

Performance optimizations include lazy loading, efficient image caching, skeleton screens, offline-first architecture, and predictive preloading.

DEMO VIDEO

CheckAR

Conclusion & Future Roadmap

CheckAR successfully combines modern UI/UX principles with intelligent automotive diagnostics — delivering a responsive, accessible, and intuitive Flutter application for both technical and non-technical users.

Phase 2: Personalization

Personalized dashboards, AI-driven maintenance recommendations, smart preferences

Phase 2: Visualization

3D dashboard visualization, real-time engine analytics, interactive repair tutorials

Scalability

Modular UI components, multi-language support, regional mechanic databases, international vehicle standards